The problem in the Data Science world

1. There is a lack of trust from the C suites(CEO,CFO,CTO etc)/shareholders/management towards the judgement or results shown by the Data Scientist.

2. As a Data Scientist/Data Analyst, your job is not only to make nice visualizations/charts/graphs/plots, although a lot of the times, you'll be doing this anyway in your effort to explain to the upper management.

3. A lot of CEOs, CFOs complain that 'the data/analysis does not solve a business problem'

4. Then you ask, "what is considered a business problem?"
 a. Make MORE PROFIT (Increase sales, Reduce Cost)
 b. Make MORE PROFIT faster (Reduce waste)

More companies seems to find Data Scientist role to be difficult understand....

What is the ANALYSIS?

As a Data Scientist/ Data Analyst

1. Given a dataset , what kind of business question can you ask/problems can you solve?
For example: You are given a salary dataset. What kind of question can you ask/ problems can you solve.

(A) As a CEO, I would be interested in how to pay my employee as competitively as legally possible in the market? (Reduce cost: Human resources, Increase Profit: Good talents can bring good profit)

2. Given a dataset, what kind of insights and analysis can you derive?
(A) What are the factors most correlated with salary?On average, which country earns the most?

3. If you are given a problem to solve (by the C suites), what kind of data that you need?

CEO:"Hey, I want to attract good talents in Sales department, but I don't to burn my cash to hire them. Can you help? "

You: "Ok, I'll need a dataset of salary ranges of our companies and other companies as well (the market). So I need the salary dataset"

Analysis result 1.
The average salary that a sales person makes in the company : USD5000
The average salery that a sales person makes in the market : USD 5500
Conclusion : Your company is paying too low. You won't get best talents

Analysis result 2.
The average salary that a sales person makes in the company : USD 5500
The average salery that a sales person makes in the market : USD 5500
Conclusion: Your company matches the market. But this wont attract good/best talents. You want to pay a little bit more "above average" say USD 5750

How to look at a dataset

	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value
2960	-117.77	33.53	32.0	3116.0	661.0	1105.0	543.0	5.1837	445600.0
7059	-118.32	34.11	42.0	2462.0	543.0	857.0	482.0	4.0833	434400.0
6101	-118.23	34.14	25.0	2864.0	844.0	1745.0	803.0	2.9167	224300.0
3438	-117.89	33.86	28.0	1395.0	398.0	1220.0	362.0	3.3008	193800.0
3917	-117.96	34.09	6.0	1954.0	534.0	1584.0	496.0	3.1621	131000.0

Given the house pricing dataset above, what is the first thing that you will do ?

How to look at a dataset

[features/attributes/ input variables / input dimension]

feature 1 feature 2 feature x_8 target output

	longitude (x_1)	latitude (x_2)	housing_median_age (x_3)	total_rooms (x_4)	total_bedrooms (x_5)	population (x_6)	households (x_7)	median_income (x_8)	median_house_value (y)
2960	-117.77	33.53	32.0	3116.0	661.0	1105.0	543.0	5.1837	445600.0
7059	-118.32	34.11	42.0	2462.0	543.0	857.0	482.0	4.0833	434400.0
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5 x 8 matrix

5 x 1 col vector

$x = 1$, scalar

$\mathbf{x} = [1\ 2\ 3]$ row vector (1×3)

$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ column vector (3×1)

$$\mathbf{X} = \begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} \\ \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \end{array} \quad (3\times3 \text{ matrix})$$

	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	y
	longitude	latitude	housing_median_age	total_rooms	total_bedrooms	population	households	median_income	median_house_value
2960	-117.77	33.53	32.0	3116.0	661.0	1105.0	543.0	5.1837	445600.0
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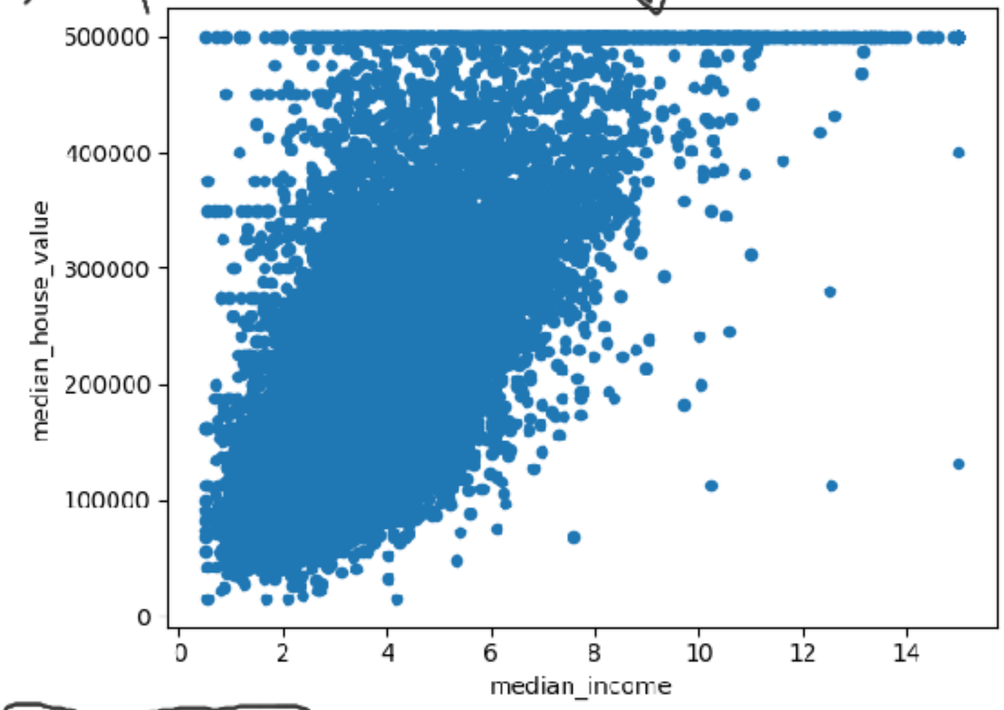
y : dependant variable

$$X = \begin{bmatrix} x_1 & x_2 & \cdots & x_8 \\ & & & \\ & & & \end{bmatrix}$$

$$x_1 = \begin{bmatrix} -117.77 \\ \vdots \\ \vdots \end{bmatrix} (17000 \times 1)$$



$\boxed{\text{other features}}$ = x_8 : Independent variable

What to check when you have a dataset?

Q1) Does the dataset have enough rows and columns

(100 rows × 4 is very small)

(1000 rows × 8~10 is ok)

(10,000 rows, 40 columns ... good)

Q2) The ratio of rows and columns (ill inverse problem)

(a) Too many rows, few columns

[ ] 10,000 × 3

: Too few features

(b) Too many columns, few rows

[ ] 15 × 120

Too many features, too few samples.

What to check when you have a dataset

Q4) Null data

Q5) Missing data.

Q6.) Corrupted data.

What is the difference between

categorical vs non-categorical variable.

1

Features 20 ~ 100

Attractive

Kind	Height	weight	humour	CGPA	salary	Attractive level
						97
		-	-	-	-	54

1000 rows / samples